# Homework Assignment 5

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**1.** (2D.2) Prove that there exists a bounded set $A \subset \mathbb{R}$ such that $|F| \leq |A| - 1$ for every closed set $F \subset A$.

**2.** (2D.5) Prove that if $A \subset \mathbb{R}$ is Lebesgue measurable, then there exists an increasing sequence $F_1 \subset F_2 \subset \cdots$ of closed sets contained in $A$ such that

$$\left| A \setminus \bigcup_{k=1}^{\infty} F_k \right| = 0.$$

*Proof.* By Theorem 2.71, we can choose closed sets $G_1, G_2, \dots$ contained in $A$ such that $|A \setminus \bigcup_{k=1}^{\infty} G_k| = 0$. We want the sequence to be increasing, so define $F_n = \bigcup_{k=1}^{n} G_k$. Then $F_n \subset F_{n+1}$ for all $n$, and each $F_n$ is closed since it is a finite union of closed sets. Since $\bigcup_{k=1}^{\infty} F_k = \bigcup_{k=1}^{\infty} G_k$, we have

$$\left| A \setminus \bigcup_{k=1}^{\infty} F_k \right| = \left| A \setminus \bigcup_{k=1}^{\infty} G_k \right| = 0.$$

□

**3.** (2D.7) Prove that if $A \subset \mathbb{R}$ is Lebesgue measurable, then there exists a decreasing sequence $G_1 \supset G_2 \supset \cdots$ of open sets containing $A$ such that

$$\left| \left( \bigcap_{k=1}^{\infty} G_k \right) \setminus A \right| = 0.$$

*Proof.* By Theorem 2.71, there exist open sets $F_1, F_2, \dots$ containing $A$ such that

$$\left| \left( \bigcap_{k=1}^{\infty} F_k \right) \setminus A \right| = 0.$$

Similar to the proof to 2D.5, we can now define $G_1, G_2, \dots$ by $G_n = \bigcap_{k=1}^{n} F_k$. Then each $G_n$ is open and contains $A$, and $\{G_k\}$ is a decreasing sequence, so

$$\left| \left( \bigcap_{k=1}^{\infty} G_k \right) \setminus A \right| = \left| \left( \bigcap_{k=1}^{\infty} F_k \right) \setminus A \right| = 0.$$

□

**4.** (2D.9) Prove that the collection of Lebesgue measurable subsets of $\mathbb{R}$ is dilation invariant. More precisely, prove that if $A \subset \mathbb{R}$ is Lebesgue measurable and $t \in \mathbb{R}$, then $tA$ (which is defined to be $\{ta : a \in A\}$) is Lebesgue measurable.